

Predictive Modeling of PCOS Using Logistic Regression: Incorporating Hormonal, Clinical, and Lifestyle Parameters

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Abstract

Polycystic Ovary Syndrome (PCOS) is a common endocrine disorder among reproductive age women, characterized by hormonal imbalances, irregular menstrual cycles, and metabolic challenges. This study investigates the effectiveness of a logistic regression model in predicting the presence of PCOS using lifestyle, clinical, and hormonal attributes. After comprehensive preprocessing and exploratory analysis—including correlation matrices and heatmaps—a logistic regression model was developed and evaluated. The model achieved an accuracy of 89.8%, demonstrating a strong potential for clinical applicability in early PCOS screening.

Introduction:

Polycystic Ovary Syndrome (PCOS), also known as Stein-Leventhal syndrome, affects 8–13% of women of reproductive age. It is associated with symptoms such as irregular cycles, obesity, hirsutism, acne, and infertility, significantly affecting quality of life. Early diagnosis is crucial to mitigate long-term comorbidities such as diabetes and cardiovascular disease. This project focuses on building a predictive model for Polycystic Ovary Syndrome (PCOS) using logistic regression, a widely used statistical method for binary classification. PCOS is influenced by a combination of hormonal imbalances, metabolic irregularities, and lifestyle factors, making it a complex condition to diagnose early. By leveraging clinical, lifestyle, and biochemical features, this study aims to estimate the likelihood of PCOS occurrence in women of reproductive age. Logistic regression was selected for its simplicity, transparency, and interpretability—qualities that are particularly valuable in clinical settings, where understanding the influence of each predictor on the outcome can support informed decision-making by healthcare professionals.

Dataset Description:

The dataset was sourced from Kaggle and comprises features such as:

- **Clinical:** Cycle irregularity, weight gain, skin darkening
- **Hormonal:** AMH levels, insulin
- **Lifestyle:** Fast food consumption

Other: Hair growth, follicle count (left/right), pimples

Data preprocessing included handling missing values, converting categorical responses to binary format, and normalizing where necessary. The target variable is binary: presence (1) or absence (0) of PCOS.

Data Visualization:

1. **Boxplots for Numerical Variables:** Features like Weight, BMI, Hb, and Pulse Rate show clear outliers. Age and BMI are moderately skewed.
2. **Histograms:** Age, Weight, and Height show near-normal distribution. Hormonal features like AMH, LH, and TSH are heavily skewed.
3. **Grouped Bar Charts:** PCOS-positive patients have higher Weight, BMI, Pulse Rate, and AMH levels. Blood group shows minimal difference.
4. **Count Plots for Binary Symptoms:** Hair growth, skin darkening, and weight gain are more frequent in PCOS-positive women.
5. **Correlation Heatmap (Top Features):** Follicle count (right and left), skin darkening, and hair growth are strongly correlated with PCOS.

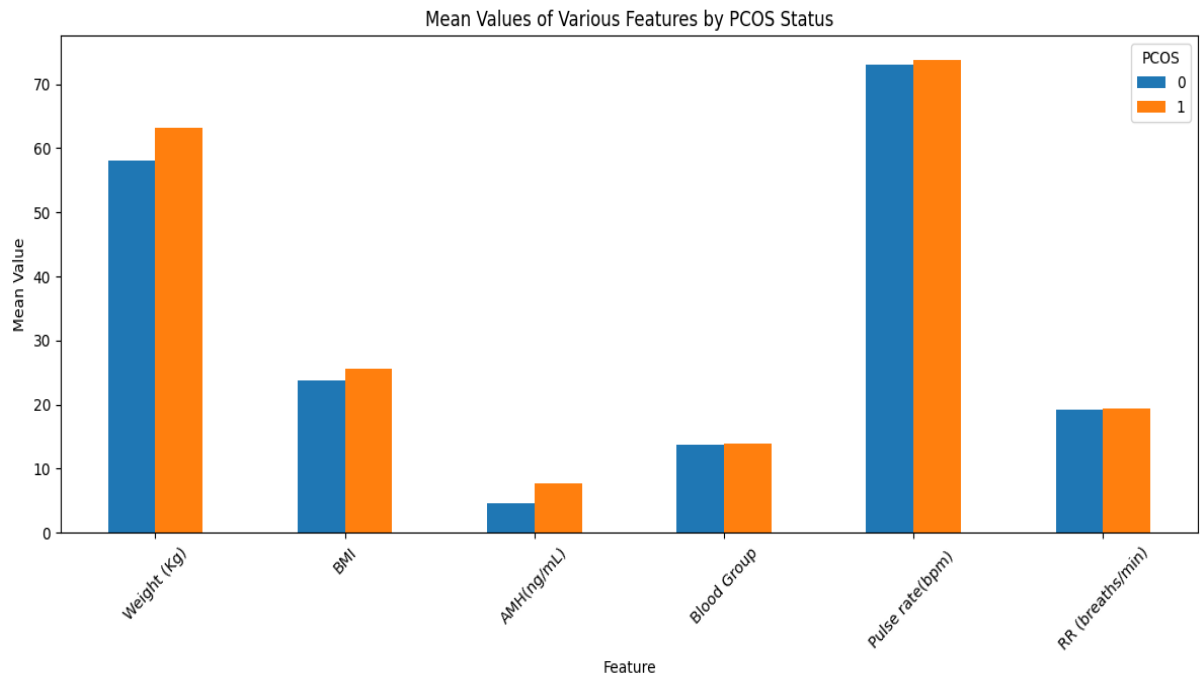


Figure 1

Distribution of Numeric Features

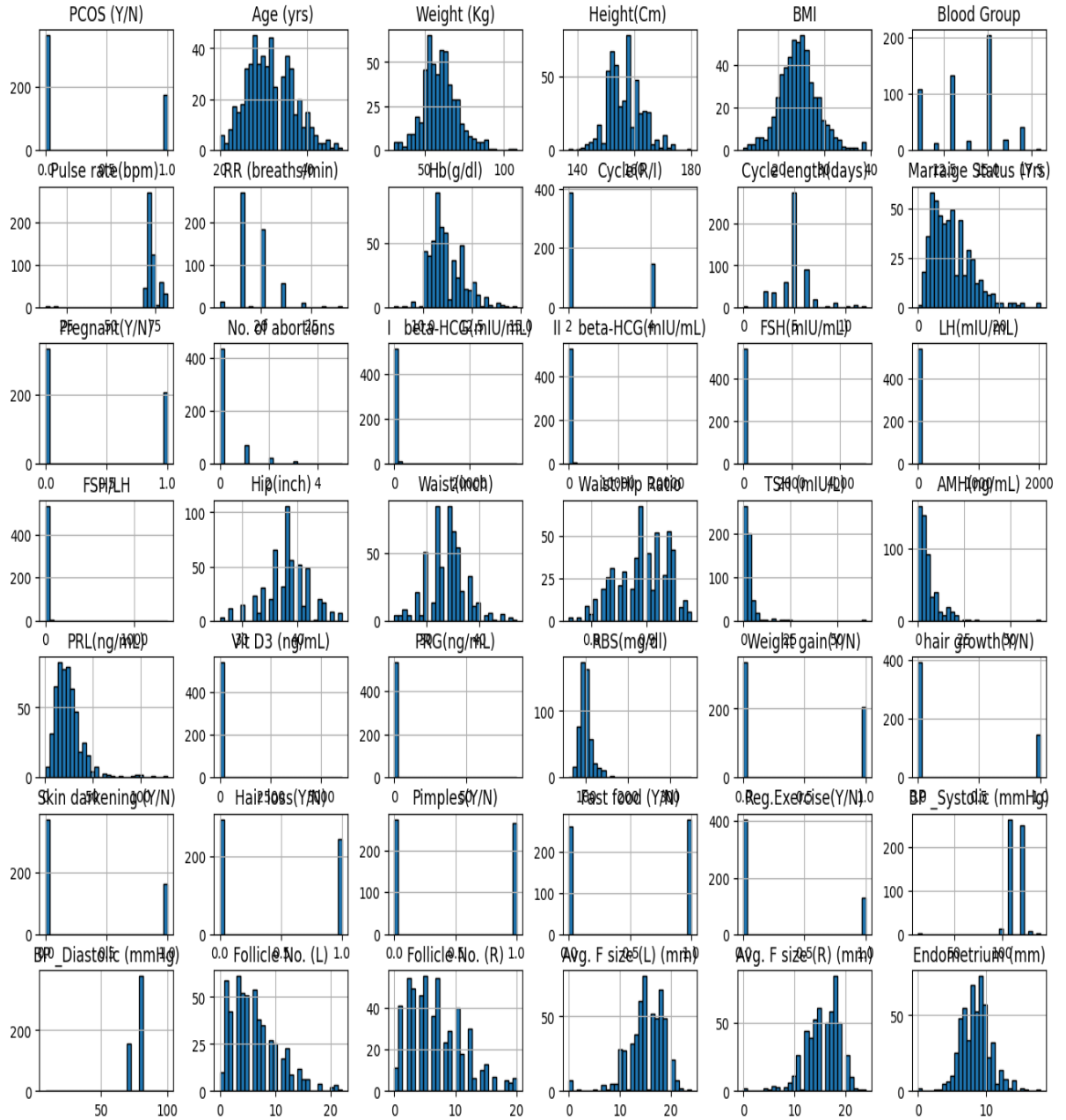


Figure 2

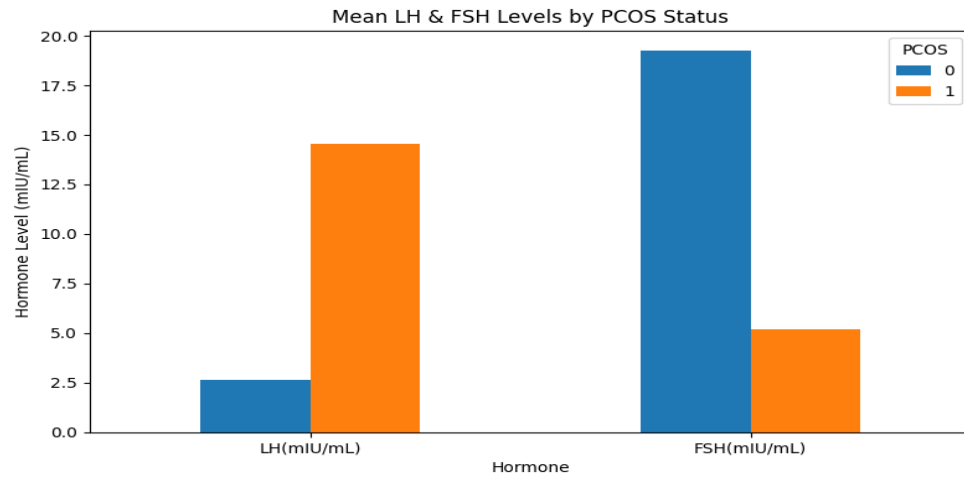


Figure 3

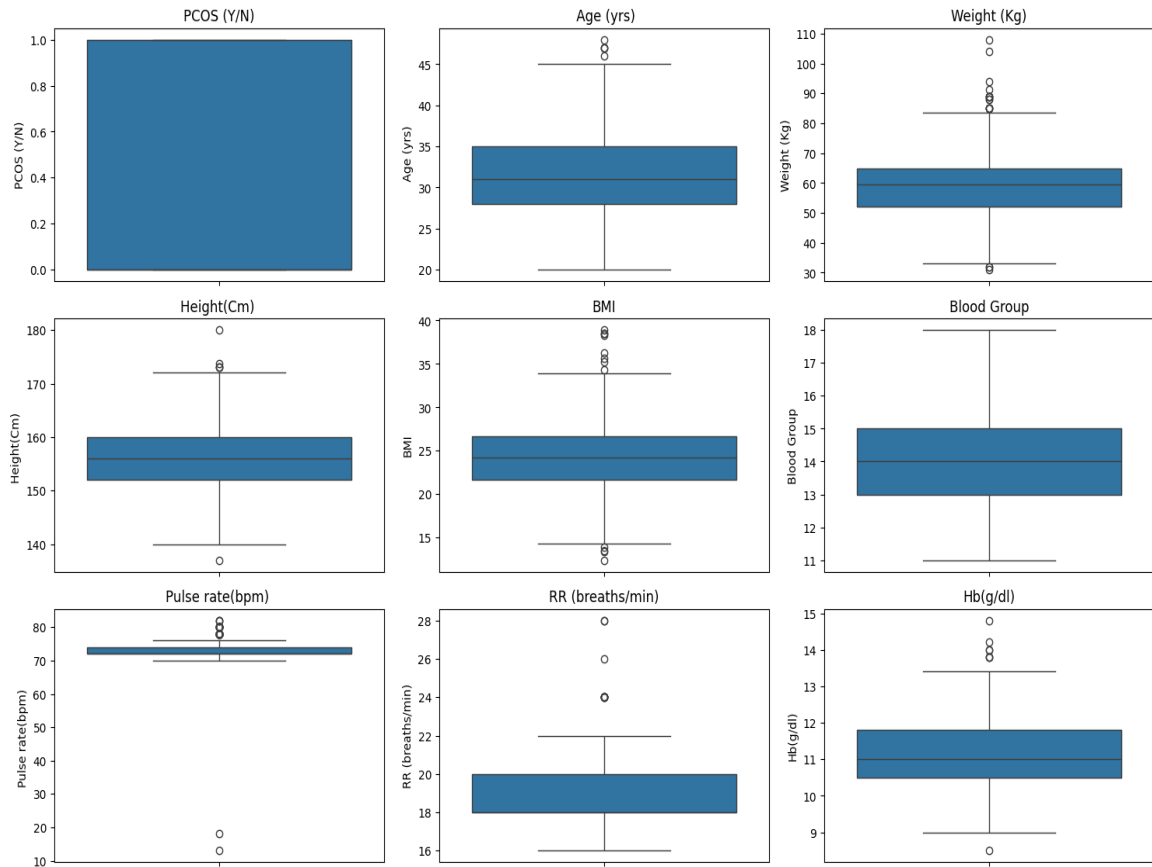


Figure 4

Feature Observations:

We generated a correlation matrix and heatmap to identify strongly associated predictors of PCOS. Features such as AMH, weight gain, irregular cycles, and insulin levels showed notable correlation.

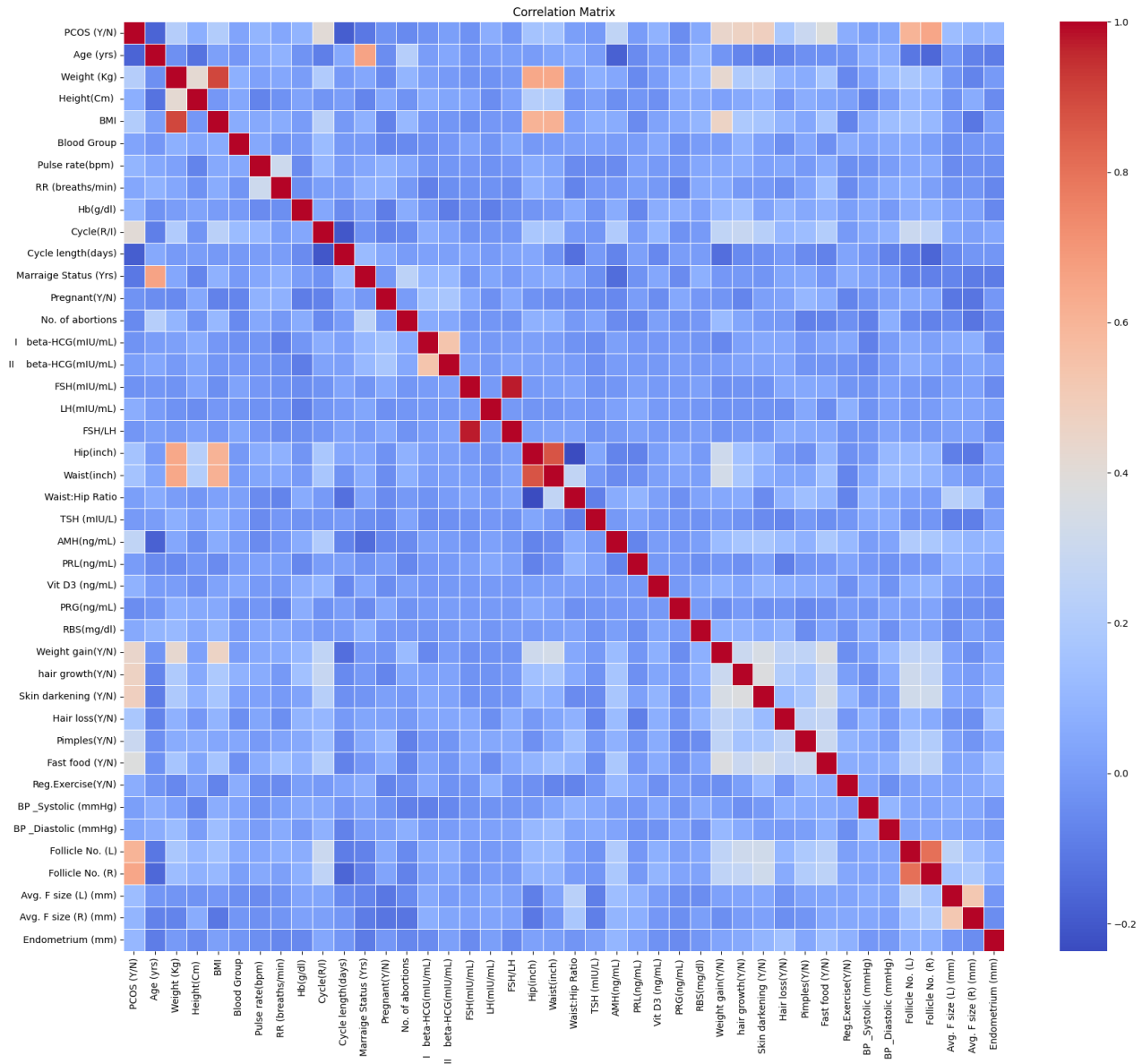
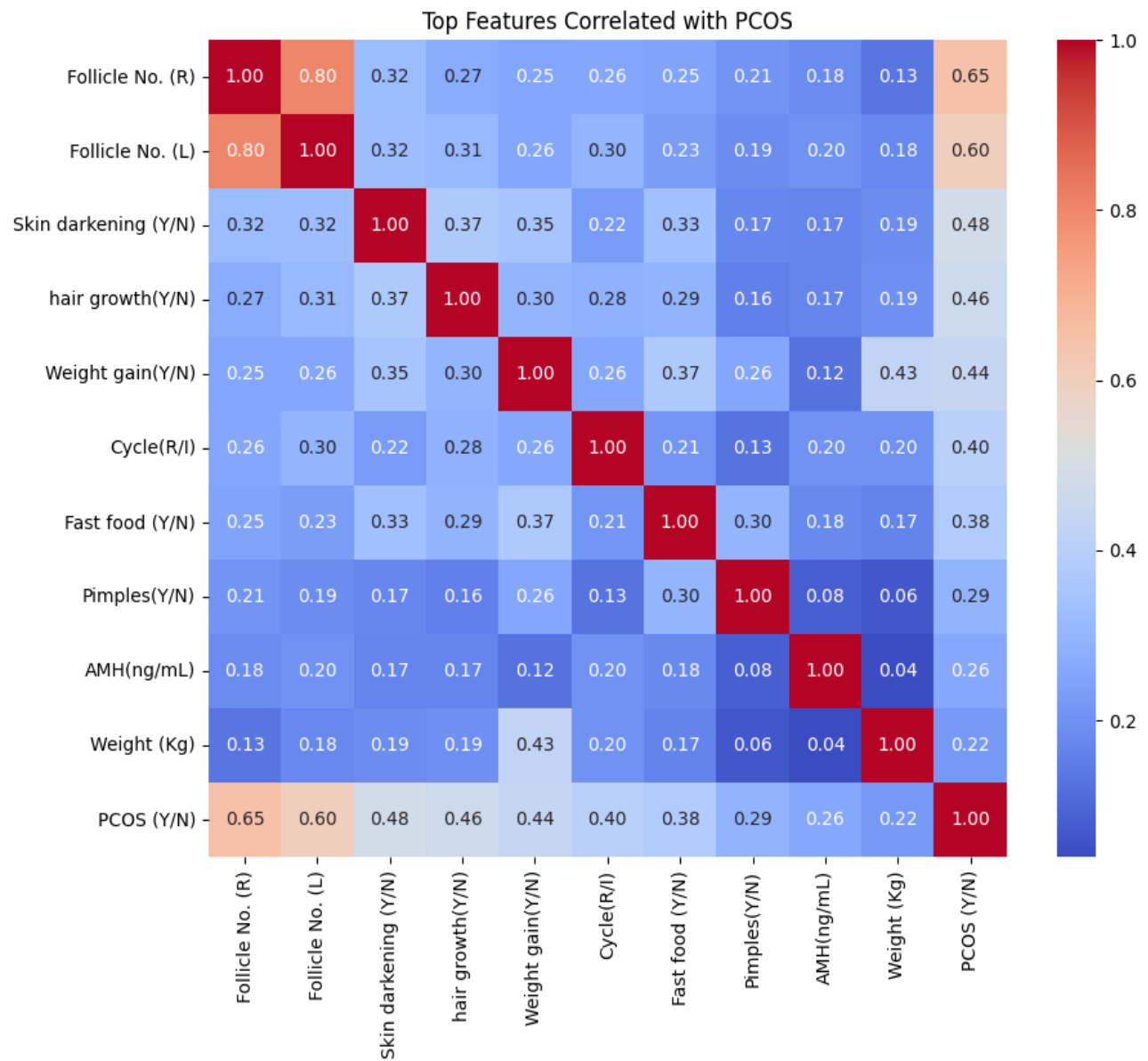


Figure 5

Figure 6: Correlation Heatmap of Selected Features heatmap



Feature	Observation	Potential Im
Follicle No. (R/L)	Strong correlation with PCOS (0.65, 0.60)	Very High
Skin Darkening, Hair Growth, Weight Gain	Strong categorical patterns by PCOS	High
AMH (ng/mL)	Higher in PCOS	High
Weight, BMI, Pulse	Slightly higher in PCOS	Medium
Blood Group	No difference	Likely Irrelevant
Cycle Irregularity	Linked with PCOS	Important
Distributions	Skewed hormone features	Requires prep

Model: Logistic Regression

Our main goal here is to obtain suitable model that captures all the optimum factors and uses the optimum. A logistic regression model was trained using 80% of the data and tested on the remaining 20%. The logistic model outputs probabilities indicating the likelihood of PCOS. The general form of the logistic function is:

$$P(y = 1|X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \dots + \beta_n x_n)}}$$

Model coefficients indicate the weight of each feature. Positive coefficients increase the probability of PCOS, while negative ones reduce it.

Evaluation Metrics

The performance of the model was evaluated using accuracy, confusion matrix, and precisionrecall metrics.

- **Accuracy:** 89.8%
- **Confusion Matrix:**

	Predicted No	Predicted Yes
Actual No	70	4
Actual Yes	7	27

- **Precision (Class 1):** 0.87
- **Recall (Class 1):** 0.79
- **F1-score (Class 1):** 0.83

Comparison with Linear Regression

A multiple linear regression model was initially attempted but yielded poor performance:

- **MSE:** 9932.15
- **R²:** -161.53

This indicates the unsuitability of linear regression for a binary classification task. Logistic regression, with an accuracy of nearly 90%, was more appropriate.

Conclusion

This study highlights the effectiveness of logistic regression in predicting PCOS using lifestyle and hormonal indicators. The model offers clinical interpretability and strong accuracy, making it suitable for early-stage PCOS screening. This project successfully demonstrates the use of logistic regression as a practical and interpretable method for predicting Polycystic Ovary Syndrome (PCOS) based on clinical, hormonal, and lifestyle parameters. With a prediction accuracy close to 90%, the model effectively identifies key indicators such as AMH levels, follicle count, cycle irregularity, and symptoms like weight gain and hair growth.

While predictive modeling can aid in early diagnosis, managing PCOS goes far beyond algorithms. It requires a holistic, long-term strategy focused on both physical and emotional well-being. Lifestyle changes, such as maintaining a balanced, low-sugar diet, engaging in regular physical activity, and managing stress play a central role. Medical treatments like hormone therapy or insulin sensitizers may also be necessary depending on individual symptoms.

PCOS is a manageable condition, not a life sentence. With early detection, awareness, and consistent lifestyle adjustments, most women can effectively control symptoms, improve fertility, and reduce long-term health risks. This study reinforces the importance of combining data-driven tools with patient-centered care to empower women in their journey toward better reproductive and overall health.

References

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